

PARTICULAR SPECIFICATION P3

Civil and Building Works

CONTENTS

P3.1	Civil and Building Works.....	1
P3.1.1	General	1
P3.1.2	Overview of Civil and Building Works.....	1
P3.1.3	Method Statements and Submissions.....	2
P3.2	ERSS and Preparatory Works	5
P3.2.1	General Requirements.....	5
P3.2.2	Qualified Person	6
P3.2.3	Accredited Checker.....	6
P3.2.4	Preparatory Work.....	6
P3.2.5	Earth Retaining or Stabilising Structures (ERSS)	7
P3.2.6	Engagement of Qualified Person(s) for ERSS Works	9
P3.2.7	Instrumentation and Monitoring Works for ERSS.....	9
P3.3	Temporary Work other than ERSS	12
P3.3.1	Scaffolding and Temporary Support Structures	12
P3.3.2	Site Platform Levels	13
P3.3.3	Engagement of QECP and ECMO.....	13
P3.3.4	Explosive Ordnance Scan and Detection Prior to Commencement of Earthworks.....	14
P3.4	Piling.....	14
P3.4.1	Piling Work.....	14
P3.5	Concrete and structural steel Works	16
P3.5.1	Concrete Corrosion Protection.....	16
P3.5.2	Electrical and ICA Manholes and Duct Banks	16
P3.5.3	Inspection Chambers and Washout Chambers	17
P3.5.4	Digestion Pre-treatment Facility (WSD11)	17
P3.5.5	Sidestream Nitrogen Removal (WST02)	18
P3.5.6	Pipe Rack Structural Steelworks (WSW41).....	18
P3.5.7	Storm Water Drain Interface	19
P3.5.8	Benching for Flow Channels	19
P3.5.9	Provision of Anchor Points for Maintenance Works	19
P3.5.10	Miscellaneous Work	20
P3.6	Water Retaining Structures.....	21

P3.6.1	Control of Shrinkage	21
P3.6.2	Joints in Concrete.....	21
P3.6.3	Casting Walls.....	21
P3.6.4	Casting Sequence of Concrete Floors	22
P3.6.5	Imperfections in Internal Concrete Surfaces	22
P3.6.6	Protection of Concrete Floors.....	22
P3.6.7	Basement and Other Below Ground Structures	22
P3.7	Building in of Pipework and Other Items	22
P3.7.1	Reinforced Concrete	22
P3.8	Other Structures.....	24
P3.9	Road and Footpath Works.....	27
P3.10	Submissions and Coordination	28
P3.10.1	Submittal Requirements	28
P3.10.2	Works Affecting Existing Building and Structures.....	30
P3.10.3	Requirements for Coordination and Liaison with Other contractors, Government Agencies, Statutory Boards, Utilities Companies, etc.....	32
P3.10.4	Survey Coordination	32
P3.10.5	Sharing of Work Access.....	32
P3.10.6	Existing Utilities.....	32
P3.11	Earthworks.....	33
P3.11.1	General Requirements for Earthworks	33
P3.11.2	Earthworks to support the Permanent Works.....	33
P3.11.3	Fill and Backfill	34
P3.11.4	Stockpiling	34
P3.11.5	Disposal of Unwanted Materials.....	35
P3.12	Measurement of Rock Excavation	39
P3.13	Appendix.....	39

P3.1 CIVIL AND BUILDING WORKS

P3.1.1 General

- (a) This section of the Specifications relates to all civil and building Works required under the Contract. The particular requirements shall complement the general requirements in the General Specification G3 Civil and Building Works.

P3.1.2 Overview of Civil and Building Works

- (a) This section describes the minimum requirements for civil and building works. This Specification shall be read in conjunction with all other sections of the Contract.
- (b) The Contractor shall submit shop drawings, material catalogues and method statements for all civil, structural and building works to S.O.'s approval before carrying out the works at site.
- (c) The Contractor shall carry out all necessary piling, reinforced concrete works and structural steel work in accordance with the Specifications and Drawings. The Contractor may refer to the General Notes in Volume 6 as design references. The following clauses describe the main areas of work.
- (d) The civil and building works required for Contract 4B shall include and not limited to the following, which includes piling, substructure and superstructures:
 - (i) Digestion Pre-treatment – WSD11;
 - (ii) Sidestream Nitrogen Removal – WST02; and
 - (iii) Pipe Rack – WSW41.
- (e) Temporary works associated with the Earth Retaining and Stabilising System (ERSS) for the construction of underground structures, pipelines and manholes for the buildings.
- (f) Road, footpaths, surface water drainage works, landscaping and external works.
- (g) Electrical and ICA duct banks and manholes.
- (h) Interface connections with C1A, C2A, C3A, C3B1 and C4A including piling, equipment plinths, steel access platforms, staircase, pipes and pipe racks works.
- (i) Connection to sewer, storm drain, access road, electrical and ICA manholes.
- (j) Interface with equipment supplier on the delivery of the civil and building works and integrate the requirements for installation of plant and equipment to be carried out.

- (k) Coordination with architectural, building services, process, mechanical, electrical and instrumentation, controls and automation (MEICA) works to ensure all the requirements of the various disciplines are integrated in shop drawings prior to construction.
- (l) The design shall also consider future equipment described in Particular Specification P4 Process Mechanical Works.

P3.1.3 Method Statements and Submissions

- (a) The Contractor shall submit detailed method statements, design calculations and other supporting documents to the S.O. before carrying out any works at site. The requirement for these method statements is detailed in the Specifications and is summarised in the list below. The S.O. may also request additional method statements or supporting documents on any aspect of the Works where further clarification is required. The Contractor shall provide these for approval prior to the commencement of relevant construction work. Approval of a method statement by the S.O. will in no way relieve the Contractor of responsibility for the successful completion and functionality of the Works in compliance with the Specifications. There shall be no additional cost to the Board due to the Contractor providing additional method statements or supporting documents.
 - (i) Summary of method statements and supporting documents required:
 - 1) General
 - a) Concrete mix design;
 - b) Concrete coating design and application; and
 - c) Site set down areas.
 - 2) Excavation and Construction Works
 - a) Proposed sequencing of works, construction equipment and locations for temporary support facilities;
 - b) Proposed ERSS with calculations and drawings which are approved by Building and Construction Authority (BCA); and
 - c) Method statements for safe access, excavation and disposal, concreting works and installation and protection of equipment.

(b) Pre-Construction Survey

- (i) Before the Contractor commences any section of the Works the Contractor shall undertake a pre-construction survey. The Contractor shall advise the S.O. at least 48 hours in advance of his intention to undertake the pre-construction survey and shall provide the S.O. with every facility necessary to confirm that the pre-construction survey is being carried out accurately. The survey shall include but not be limited to:
 - 1) Topographical survey of the proposed working areas to confirm existing ground levels, the locations and levels of all structures and other surface features;
 - 2) Structural survey of all existing structures that may be affected by the proposed Works. The survey shall identify significant cracks and defects on structures and measure their width and length. Where the Contractor's Professional Engineer deems it necessary to fix strain markers on cracks to measure any widening of the crack width or shearing of the structure, the Contractor shall coordinate with the S.O. and submit the instrumentation and monitoring locations for the Instrumentation and Monitoring contractor to fix the crack meters; and
 - 3) Road and access track condition survey. The Contractor shall make a detailed photographic record of all roads and access tracks that will be used during the construction work and shall make careful notes of any Defects. These photos shall be copied to the S.O..
- (ii) The pre-construction survey shall include confirmation of position and level information included in the Drawings and determination of the presence, dimensions and level of all utilities, pipelines, drains, cables, chambers, structures and the like, which may be affected by the Works. In the event that the pre-construction survey reveals any significant discrepancies from the Drawings, the Contractor shall inform the S.O. immediately of all such discrepancies. At all times the Contractor shall work to the instructions of the S.O. regarding any modification to the design of the Works made necessary by the identification of such discrepancies.

(c) Post Construction Survey

- (i) Structure condition survey
 - 1) The Contractor shall undertake a comparative survey with the S.O. to determine the structural condition of all existing structures recorded in the pre-construction survey. The aim of the survey will be to identify any deterioration in the structure condition that cannot be attributed to normal operational wear.

(ii) Road and access way condition

- 1) The Contractor shall undertake a comparative survey with the S.O. to determine the condition of all roads and footpaths recorded in the pre-construction survey. The aim of the survey will be to identify any deterioration in the roads and footpaths condition that cannot be attributed to normal operational wear.

(iii) The Contractor shall repair all deterioration identified in this survey to the satisfaction of the S.O. at his own cost.

(iv) Shoring System for Deep Excavations

- 1) The Works require deep excavations to be carried out for structures and for pipelines. In advance of undertaking any such excavation in excess of 1.5 metres depth the Contractor shall submit a detailed method statement to the S.O. for approval to indicate the proposed method for supporting the excavation. The detailed method statement shall include drawings and calculations certified by a Professional Engineer, Civil and Geotechnical (Singapore), and approved by BCA.
- 2) The Contractor's attention is drawn to the proximity of existing services and drainage installed by Other contractors to the proposed construction works. The Contractor shall be required to liaise with the Other contractors and adequately protect during the construction works.

(v) Prevention of Flotation

- 1) The Contractor is warned that underground structures may not be heavy enough during construction to resist uplift due to groundwater water pressure.
- 2) The Contractor shall make adequate arrangements to keep excavations dry or adopt other Approved means of ensuring the stability of the Temporary or Permanent Works until they are completed.
- 3) The Contractor's proposed method specific to each relevant structure shall be submitted to the S.O. for approval before work commences. The Contractor shall provide for adequate standby Plant and attendance throughout the period that the structure is at risk from uplift.

(vi) Ground Water Conditions

- 1) The ground water conditions are indicated in the attached Soil Investigation Report. The Contractor shall take note of the sulphate content of the ground water and shall ensure that the concrete mix design is fully compliant with the corresponding severity class as detailed in the structural drawings.

(vii) Maximum Permissible Loadings on Roofs and Suspended Slabs

- 1) The Contractor shall limit the laden weight of his construction Plant permitted to be driven on roofs and suspended slabs and shall submit to the S.O. for approval details of all Plant, which are proposed for use on roofs or suspended slabs prior to first use. Unapproved Plant shall not be permitted to drive on roofs or suspended slabs under any circumstances. The Contractor shall not stack soil, free draining material or other construction material on roofs or suspended slabs except for the purpose of spreading and levelling the said materials thereon.
- 2) The maximum permissible imposed uniformly distributed load during construction activities for all concrete suspended slabs and roofs shall not exceed the value shown in loading plan provided in Volume 6.

(viii) Backfilling around Structures

- 1) The Contractor shall take particular care during the construction of those structures which are founded at more than one level to ensure that no foundations of any structural elements are inadequately founded as a result of the Contractor's Temporary Works or method of working.
- 2) Engineering Fill shall be used at all such locations and at any other location where components of structures are subsequently to be founded on backfilled or filled areas. Engineering Fill used as backfill shall not be measured, unless indicated otherwise in the Schedule of Prices, and no additional payment shall be made for such backfill.

P3.2 ERSS AND PREPARATORY WORKS

P3.2.1 General Requirements

- (a) The Contractor shall ensure the safety of all parties, including workers and Other contractors working in the vicinity. Proper safety barricades and adequate lighting must be installed in the work sites.
- (b) The Contractor shall maintain the work sites to be clean and tidy and ensure the work site is free from stagnation of water.
- (c) The Contractor shall engage Qualified Person(s) for the design, supervision and construction of ERSS Works as specified in Particular Specification P2 Preliminaries and General, including liaising with Accredited Checker and obtaining approval from the BCA or any other Statutory Bodies.
- (d) The Contractor shall engage Qualified Erosion Control Professional (QECP) to design, submit and obtain approval for proposed Earth Control Measures (ECM) as specified in Particular Specification P2 Preliminaries and General.

- (e) The Contractor shall provide all temporary diversions (such as pipes, drain, cables, footpath, traffic control, lamp post, road diversions, etc.) to facilitate the execution of the Works.
- (f) The Contractor shall carry out at his own expense, all remedial works to existing facilities, due to damage caused by the Contractor.
- (g) The Contractor shall provide external perimeter hoarding encompassing the entire C4B working area including the stockpile and laydown. He shall submit the P.E. endorsed design of the hoarding structures to S.O. and obtain his approval, prior to its implementation at site.
- (h) The Contractor shall note that the boundary of the perimeter hoarding may be required to be shifted during the Contract period. The Contractor shall allow in his rates and prices for the establishment and shifting within the Contractor's staging area and working areas as required by the S.O..
- (i) The Contractor shall ensure that his methods of working cause minimum disturbance to the construction activities of Other contractors.

P3.2.2 Qualified Person

- (a) The Contractor shall engage a QP(D) and QP(S) for all temporary works.
- (b) For permanent works which are under the Contractors design scope as specified, the Contractor shall engage a QP(D). The QP Supervision team (including the QP(S) for the permanent elements, will be engaged by the Board.

P3.2.3 Accredited Checker

- (a) Accredited Checkers (AC) have been appointed by the Board under a separate contract as required by the Building Control Act and the Building Control Regulations to check the design carried out by the Contractor.
- (b) The Contractor and his QP shall allow for coordination with the AC to resolve all technical and submission issues for ERSS works to the satisfaction of the AC. The Contractor is not entitled to any extension of time and cost for complying with the AC's or statutory requirements and any delay in the resolution of the technical and submission issues.

P3.2.4 Preparatory Work

The Contractor shall:

- (a) Carry out pre-construction survey of the Site and existing facilities, including but not limited to the existing site revetment, roads and drainage infrastructure. Subsequently, carry out post-condition survey at end of the Contract;

- (b) Carry out site clearance, site formation / preparation works including construction of temporary access, establishment of site offices, toilet facilities, vehicular wash-bay, environmental condition monitoring facilities and temporary staging areas;
- (c) Performing services detection and trial trenches to locate the intervening services; including purchasing the plans from the Government and other Utility Agencies;
- (d) Engage Qualified Erosion Control Professional (QECP) to design, submit and obtain approval for proposed Earth Control Measures (ECM);
- (e) Set up ECM system for the work area and the site office area. Appoint ECM Officer to coordinate with QECP on the construction, improvement and maintenance of the ECM system;
- (f) Engage a licensed surveyor to set up site survey benchmark, setting out control and grid lines;
- (g) Carry out additional soil investigations (SI) works if required by the Contractor's professional engineer engaged for ERSS design; and
- (h) Carry out ordnance survey prior to excavation to detect presence of war relics.

P3.2.5 Earth Retaining or Stabilising Structures (ERSS)

- (a) The Contractor shall:
 - (i) Provide ERSS for all permanent works in accordance with his Contractor's PE design and BCA approved drawings;
 - (ii) The permanent works shall be and not limited to the Sidestream Nitrogen Removal Facility, Buried Pipework, Valve Chambers, Manholes, Drains, External Works and any other underground structures mentioned elsewhere in the Drawings and Specifications;
 - (iii) The Contractor shall undertake the design and implement ERSS scheme;
 - (iv) Design and submit the ERSS design endorsed by Contractor's PEs to the S.O. and nominated Accredited Checker for review and subsequently obtain approval from BCA prior to the start of the work;
 - (v) Upon getting BCA Approval, the Contractor shall submit a copy of approved drawings and calculations to S.O.;
 - (vi) Construct the ERSS and carry out staged excavation in accordance with the BCA approved plans and drawings, under his QP supervision;
 - (vii) Identify the type, number and location of monitoring instruments required to monitor the condition of the ERSS during the construction;

- (viii) Removal of the ERSS, when no longer required or at the completion of works; and
 - (ix) Provide adequate number of suitably qualified and experienced site supervision personnel per the requirements of the Building Control Act and Regulations who will work under his QP(S) / QP(S)-Geo for the supervision of the ERSS and temporary works and also ensure that the work is carried out in accordance with the approved drawings and specifications.
- (b) The Contractor shall provide a suitable ground water cut off system and ERSS system to prevent water ingress before proceeding with bulk excavation to formation level.
- (c) The Contractor shall allow for in his design any water recharge system as may be required by his ERSS work. The Contractor's QP shall ensure the nearby buildings or facilities are not impacted by the drawdown of ground water table arising from the ERSS work. The recharge wells shall be installed as per Contractor QP's recommendation along the existing buildings before bulk excavation starts and shall be activated upon reaching the predicted water draw down levels.
- (d) The ERSS work includes the provision of an earth retaining system, consisting of various local sub-systems each tailored to suit the purpose at each locality, excavation to formation level, progressive removal of the ERSS system and the design of suitable instrumentation and monitoring plan.
- (e) Supervision of ERSS works during Construction:
- (i) The QP engaged by the Contractor for the design of the ERSS works shall be responsible for the proper supervision and inspection of all ERSS works in accordance with the approved plans. The Contractor's QP Supervision team shall include for QP(S), QP(S)-Geo (if required as per BCA requirements), RE and RTOs.
 - (ii) The Contractor and his QPs shall extend full assistance and co-operation to the S.O. and his representatives at all times. All works shall be carried out only with the concurrence of and to the satisfaction of the S.O.;
 - (iii) The presence of the S.O. and his representatives during the execution of the Works shall not absolve the Contractor and his QPs of the responsibility of ensuring proper execution and supervision of the Works;
 - (iv) Provide adequate number of suitably qualified and experienced site supervision personnel per the requirements of the Building Control Act and Regulations who will work under his QP(S) / QP(S)-Geo for the supervision of the ERSS and temporary works and also ensure that the work is carried out in accordance with the approved drawings and specifications; and
 - (v) The Contractor's QP shall check the instrumentation and monitoring readings regularly and submit the report to the S.O. according to the regulation.

P3.2.6 Engagement of Qualified Person(s) for ERSS Works

- (a) The Contractor shall engage a Professional Engineer and a specialist Professional Engineer (Geotechnical) as qualified persons (QP) for the design, supervision and construction of ERSS works. The design and construction of the ERSS works shall comply with the requirements of the Building Control Act, the Building Control Regulations and all the relevant Authorities, and approval by the S.O.
- (b) The Contractor's QPs shall prepare and endorse the necessary calculations, details and drawings and submit it to the S.O. and AC for checking. It is the responsibility of Contractor's QP to submit the ERSS drawings and calculations to BCA to obtain the approval and permit to commence work.

P3.2.7 Instrumentation and Monitoring Works for ERSS

- (a) The Contractor shall take note that the Board appointed a specialist Instrumentation and monitoring contractor (IMC) under a separate contract to provide and maintain various instruments required for monitoring the Contractor's ERSS works based on the Instrumentation and Monitoring Plans prepared by the Contractor and accepted by the S.O.. The Contractor shall provide all necessary access and coordinate instrumentation and monitoring works with the IMC in a safe and timely manner. The Contractor shall work with the S.O. to ensure that all instruments types and locations are identified.
- (b) The Contractor shall notify the S.O. at least three (3) weeks in advance for any instrumentation that are to be installed by the IMC.
- (c) The Contractor shall design, plan and prepare detailed design drawings for instrumentation and monitoring required to carry out the Works in accordance with the Contract, Authorities and stakeholder's requirements to the acceptance of the S.O.. The Contractor shall submit the detailed design drawings for the instrumentation and monitoring works together with an Instrumentation Design Report. The Instrumentation Design Report shall describe in detail the basis for proposed instrumentation, identify, where relevant, the associated risk items in the Contractor's Risk Register, and propose additional entries into the IMC's Instrumentation Register. The Instrumentation Design Report and detailed design drawings of additional instrumentation shall be submitted at least six (6) weeks in advance of the commencement of the Works.
- (d) The Contractor shall assist the IMC for the installation of any monitoring equipment required within the Site. This shall include for temporary access for installation and monitoring and protection of instruments. The Contractor shall bear the costs for replacement of any instruments damaged by his Works.

- (e) The Building and Construction Authority (BCA) and other regulatory bodies require monitored instrumentation readings to be observed against pre-defined review levels and limits. The Contractor shall take into account all relevant Authorities' requirements in deciding the instrumentation and defining appropriate review levels. The Contractor shall take into account the requirement for installation and monitoring of instrumentation by BCA and other regulatory bodies. The Contractor, QP(S) Civil / Geo and QP(D) Civil / Geo shall review the readings regularly and take appropriate actions as required.
- (f) The design shall include tables giving reading frequencies for all instruments. The frequency of reading the instruments may be varied, depending on the type of instrument and the relationship between the instrument and areas of on-going activity. Criteria for increasing or decreasing the frequency of reading and for ceasing monitoring shall also be given, subjected to the acceptance of the S.O..
- (g) The Contractor shall note whilst the required types of instrumentations are included inside the specialist Instrumentation and monitoring contract (IMC), any special instruments that are not in the IMC shall be provided and monitored by Contractor at Contractor's own cost. The type of instruments that are included in the IMC is provided below for reference:
 - (i) Building Settlement Markers;
 - (ii) Ground Settlement Markers in unpaved ground (rod type in soil);
 - (iii) Ground Settlement Markers on concrete surface, asphalt or other pavement/rigid ground (nail type);
 - (iv) Utility Settlement Markers;
 - (v) Deep Levelling Datum;
 - (vi) Electro-level Beam Sensors;
 - (vii) Inclinometers (in soil/wall);
 - (viii) Magnetic Extensometers;
 - (ix) Rod Extensometers;
 - (x) Tape Extensometers;
 - (xi) Vibrating Wire Piezometers;
 - (xii) Water Standpipes;
 - (xiii) Heave Stakes;

- (xiv) Vibrating Wire (VW) Strain Gauges;
 - (xv) Vibration Monitoring Sensors;
 - (xvi) Load Cells;
 - (xvii) Crack Meters;
 - (xviii) Tilt Meters;
 - (xix) Optical / 3D Glass Prisms; and
 - (xx) Automated Total Stations.
- (h) Instrumentation Plans
- (i) The Contractor shall prepare instrumentation plans at a scale not greater than 1:500 and shall clearly indicate the location and installation height/depth of all instrumentation. Sectional drawings shall show the extent and depth of permanent and temporary works and the strata where the instruments are to be installed. All instruments shall be clearly identified in a unique alpha-numeric sequence. Each drawing shall show all the instrumentation to be installed within the area covered in the drawing. At contract boundaries, the instruments that are necessary to be installed outside this contract boundary shall also be shown.
 - (ii) Where multi-sensor instruments are used, they shall be clearly identified with the reduced level and/or direction of each individual sensor. Such information shall be provided on the drawings with tables.
 - (iii) Where buildings, structures or utilities subjected to special protection works are encountered, a detailed instrumentation drawing shall be prepared for each individual building, structure or utility indicating the exact location of all monitoring points, including the reduced level of the proposed instrument(s). Sectional drawings shall also be prepared showing clearly the extent of any existing features, cracks etc. identified from the Contractor's Pre-Construction Survey which require monitoring. Where possible, the structural layout of the building shall also be indicated on these drawings.
 - (iv) All drawings for buildings, structures or utilities which are subjected to special protection works shall be prepared at a scale not greater than 1:50 except where prior acceptance has been granted by the S.O..
- (i) Instrumentation Tables
- (i) Instrumentation tables shall include but not be limited to the following information for each instrument, subjected to the acceptance of the S.O.:
 - 1) Instrument type;
 - 2) Unique instrument number;

- 3) Sensor number;
- 4) Proposed easting;
- 5) Proposed northing;
- 6) Proposed reduced level of all sensors;
- 7) Proposed reading frequency during key stages of the Works; and
- 8) Review levels for all sensors.

P3.3 TEMPORARY WORK OTHER THAN ERSS

P3.3.1 Scaffolding and Temporary Support Structures

In General, the Contractor shall,

- (a) Emphasise safety, speed and ease of operation with minimum manual handling operations.
- (b) Look at the ease and speed at which the forms can be stripped, cleaned and re-set as well as minimising intermediate storage at site.
- (c) Carry out risk assessment for formwork and platform operations.

The Contractor shall:

- (a) Provide PE endorsed design of the following;
 - (i) Scaffolding systems for man access during construction;
 - (ii) Scaffolding system for supporting formwork; and
 - (iii) Formwork.
- (b) Provide PE endorsed design and certificate of supervision for miscellaneous temporary supports, work platforms, etc.
- (c) Propose the technique of jump form used on the cylindrical wall construction shall permit continuous climbing without intermediate storage of formwork panels at ground level required. The following features shall be included:
 - (i) The technique of jump form used on the cylindrical wall construction shall permit continuous climbing without intermediate storage of formwork panels at ground level required;
 - (ii) At the construction joint formwork from the previous cast section shall remain on the structure to eliminate mortar leakage;

- (iii) Self-supporting formwork system without the requirement of through-wall form ties;
 - (iv) Low number of bolt connections for easy and fast erection and dismantling works;
 - (v) Formwork shall be constructed from members curved in plan to match the radius of the curvature of the wall. Most preferably laser cut prefabricated large elements for easy and fast construction;
 - (vi) Working platforms inside and outside shall be equipped with hand, knee and toe rail, access doors and platforms preventing any objects from falling. Especially toe rail and platform shall not allow any small objects from falling (caged system); and
 - (vii) No jump form system that permits leakage through holes in the forms, between the forms, or between the forms and previously placed concrete shall be used.
- (d) Propose and construct the site wide drainage system for the interception, control and management of surface runoff, collection and discharge of water from direct rainfall.
- (e) Ensure that temporary works or any other design works designed by Contractor's PE shall be supervised by him during construction and certificate of supervision shall be submitted to the S.O. upon the completion of these works.

P3.3.2 Site Platform Levels

- (a) The Contractor shall be responsible for providing and maintaining Site wide construction access for his working areas and working platform levels. The Contractor shall provide safe construction access by means of ramps to each interim formation level associated with each stage of work. The cost for the provision of access to his work areas and working platforms shall be deemed to be within the rates and prices.

P3.3.3 Engagement of QECP and ECMO

- (a) The Contractor shall engage a qualified QECP to design and make submission to PUB C&W department to obtain approval for the proposed Earth Control Measure (ECM) for the site. The QECP shall make expeditious submission to approving authority to ensure no delay to the project execution.
- (b) The Contractor shall appoint or engage a qualified ECM Officer (ECMO) for the project. The ECMO shall implement the ECM measures according to QECP plan, devise ECM operations, set up maintenance and inspection programme, and assist the QECP to monitor ECM effectiveness throughout the various stages of work.

P3.3.4 Explosive Ordnance Scan and Detection Prior to Commencement of Earthworks

- (a) The Contractor shall engage the services of an Unexploded Ordnance (UXO) service provider to carry out scanning and detection of any explosive ordnance for up to an area of one (1) hectare of the site before carrying out any excavation or trial trenching works in the vicinity of the Works. The location of the scanning will be as directed by S.O..
- (b) Shall any anomalies be found; the Contractor shall adopt the recommendations of the Unexploded Ordnance (UXO) service provider as directed by the S.O. with regards to the methodology of any excavation or trial trenching works accordingly.

P3.4 PILING

P3.4.1 Piling Work

- (a) The Contractor shall provide piling/foundations for the following structures:
 - (i) Sidestream Nitrogen Removal - WST02;
 - (ii) Others, including, but not limited to the followings:
 - 1) Pipe / Equipment Support;
 - 2) Drains;
 - 3) Manholes / Chambers; and
 - 4) Other miscellaneous structures such as cable ducts, etc.
- (b) The Contractor shall:
 - (i) Submit proposals and obtain the S.O.'s approval for:
 - 1) Carrying out the ultimate load test;
 - 2) Installing test pile(s) at agreed location(s) for the specified pile types;
 - 3) Providing the number of required ultimate pile load test(s) as specified in the Contract and carrying out ultimate pile load test based on the available SI data; and
 - 4) Preparing the cut off level of the bored piles.
 - (ii) Complete pile testing and submit ultimate pile load test report and interpretation of design parameters within 4 weeks from the commencement of the Works, with specialist PE's interpretation for the proposed design parameters, and obtain S.O.'s approval;

- (iii) Provide a full time Bored Piling Engineer (BPE) – The BPE shall be a geotechnical engineer or geologist or as approved recognized by the Professional Engineers Board of Singapore and shall be full time during the execution of bored piling works. He shall be able to interpret ground conditions and prepare geotechnical interpretative report during boring works for submission to the S.O. for approval. The BPE shall have a minimum of 5 years experience as a geotechnical engineer or geologist;
- (iv) Verify each pile capacity at the installed location, provide soil samples, classification of the type of differing soil layers and detailed boring records (complete with documented depths of the distinct soil layers/thicknesses that differ from the design bore log) of the affected piles for the QP(D) and QP(S) to review;
- (v) Submit revised pile length with PE's endorsement for each pile type based on the approved design parameters, should the soil condition encountered at Site be different;
- (vi) Confirm pile setting out with the S.O. or his representative on site;
- (vii) Install piles or any other proposed pile systems at the approved setting out location and to the required depth and penetrations;
- (viii) Verify each pile capacity at the installed location by the Contractor's PE;
- (ix) Submit the pile working load test method statement and obtain S.O.'s approval;
- (x) Carry out the following pile tests on selected piles as approved by S.O. and submit the test reports endorsed by the piling specialist PE:
 - 1) Static working load test;
 - 2) Pile dynamic analyser (PDA) test; and
 - 3) Pile integrity test (PIT).
- (xi) Carry out additional working load testing, reduce the number of tests or to vary the test requirements as directed by the S.O.;
- (xii) Prepare pile heads, carry out pile head position survey by licensed surveyor and submit report on pile eccentricity (ECC);
- (xiii) Propose and recommend any remedial work and submit accompanying design checks by a PE or a specialist PE as required by authorities for the pile affected by excessive ECC for the QP(D) to review and approve. The Contractor's PE will design the remedial measures for piles with eccentricity exceeding the tolerances given in the Drawings or Specifications; and

- (xiv) Prepare as-built piling plan, as-built piling table and BIM Model pile tagging according to BCA requirements and timeline. These are to be endorsed by PE or a specialist PE as required by authorities and licensed surveyor for submission to BCA, upon substantial completion of the works.
- (c) The design of the piling system shall be based on SS EN 1997.
- (d) The S.O. reserves the right to instruct additional working load testing, reduce the number of test or to vary the test requirements.
- (e) The Contractor's PE shall design the remedial measures for piles with eccentricity exceeding the tolerances and all the additional measures required to address the eccentricity shall be executed by the Contractor without claiming any additional cost.

P3.5 CONCRETE AND STRUCTURAL STEEL WORKS

The Contractor shall carry out all necessary reinforced concrete works and structural steel work in accordance with the Specifications and Drawings. The following clauses describe the main areas of work.

P3.5.1 Concrete Corrosion Protection

- (a) All Liquid Retaining Structures shall have corrosion protection system as specified in Appendix P3A.

P3.5.2 Electrical and ICA Manholes and Duct Banks

- (a) The Contractor shall provide electrical and ICA duct banks and manholes for underground electrical power, telecommunications and control cables from the interfaces as shown on the Drawings and Volume 6 Reference Model. Cable ducts are arranged in arrays and these vary depending on location. The locations of different arrays are shown in the Drawings. Cross sections of arrays are shown on the Drawings.
- (b) The Contractor shall note the following:
 - (i) Ducts are to be spaced as shown on the Drawings;
 - (ii) Electrical and ICA manholes shall be provided and final location shall be approved by S.O.;
 - (iii) Lead-in pipes extended from terminal manholes to interface points (TP) with Other contractors are to be protected and sealed against the ingress of water, soil and vermin;
 - (iv) Spacers shall be provided between ducts to maintain the space between conduits as required; and

- (v) The Contractor may refer to C1A and C3A Structural Standard details in Volume 6 for duct bank construction.
- (c) The Contractor shall coordinate the works with the S.O., and Other contractors for interfacing works, schedules, etc. The Contractor shall note that all programming clashes with Other contractors shall be properly coordinated and resolved. The Contractor shall note that the S.O.'s decision will be final on any programming coordination.
- (d) Refer to Particular Specification P2 Preliminaries and General, Appendix P2B for the Tie-In Interface Register detailing interface scopes and locations.
- (e) The typical cross section of duct banks shown in the Drawings is for reference only. The exact number of pipes and layers may have to be adjusted to site conditions to the S.O.'s approval where necessary.
- (f) Where there is a change in the tender configuration of the ducts, and if there is no standard typical arrangement available in the Contract, the Contractor shall proposed the suitable arrangement for S.O. approval and is deemed that these are included in the Contract Sum.

P3.5.3 Inspection Chambers and Washout Chambers

The Contractor shall provide the following:

- (a) Provide all reinforced concrete inspection chambers and washout chambers complete with covers and connections to pipes and drains as shown in the drawings;
- (b) Submit shop drawings, materials catalogues and method statements; and
- (c) Provide PE endorsed designs and obtain BCA approval (if required) for pipe supports.

P3.5.4 Digestion Pre-treatment Facility (WSD11)

- (a) Existing foundation had been designed for 80kN/m² imposed load. The imposed floor loadings shall take into consideration of the steel platform, access staircase, pipe rack support or other supporting structure, e.g. steel frame, cradle or concrete plinths etc. loads.
- (b) The Contractor shall engage a Professional Engineer as qualified person (QP) for the design and construction of the Digestion Pre-treatment Facility (WSD11). The Contractor's QP need to ensure the equipment operational, testing, and accidental load (both force and moment) at each supports point have been considered especially the dynamic factor during equipment operation to ensure the existing foundation capacity is adequate. The Contractor's QPs shall prepare and endorse the necessary calculations, details and drawings and submit it to the S.O. for checking.

- (c) It is the responsibility of Contractor's QP to submit the drawings and calculations to BCA to obtain the approval and permit to commence work where applicable.

P3.5.5 Sidestream Nitrogen Removal (WST02)

- (a) The Contractor shall design and build the Sidestream Nitrogen Removal facility (WST02) including piling foundation and superstructure works. Design responsibility for civil and structural works shall be in accordance with the Specifications. The Contractor may refer to the concrete durability requirement (Exposure Class Specification) in Volume 6 as design reference. The design is to comply with all the requirements of Eurocode and Singapore Annex. A set of Soil Investigation report is provided in Volume 6 for reference only.
- (b) The Contractor shall engage a Professional Engineer as qualified person (QP) for the design and construction of the Sidestream Nitrogen Removal facility (WST02). The design and construction works shall comply with the requirements of the Building Control Act, the Building Control Regulations and all the relevant Authorities, and approval by the S.O..
- (c) The Contractor's QPs shall prepare and endorse the necessary calculations, details and drawings and submit it to the S.O. and AC for checking. It is the responsibility of Contractor's QP to submit the drawings and calculations to BCA to obtain the approval and permit to commence work.
- (d) The Contractor's QP to cater for the 2.0kPa live load at roof level for Solar Panel installation in future.

P3.5.6 Pipe Rack Structural Steelworks (WSW41)

- (a) Below is the list of Pipe Rack Structural Steelworks (WSW41) interface connection with the other contracts:
 - (i) The pipe rack interface connection with C4A and C3B1 pipe rack;
 - (ii) The pipe rack interface connection with C1A and C2A piles foundation; and
 - (iii) The pipe rack interface connection with C2A WSE21 Electrical Main Substation B Building.
- (b) The Contractor shall design and build the pipe rack including their interface connections. A notional scheme is described in the Drawings. The Contractor shall engage a Professional Engineer as qualified persons (QP) for the design, supervision and construction. The design and construction shall comply with the requirements of the Building Control Act, the Building Control Regulations and all the relevant Authorities, and approval by the S.O.
- (c) The Contractor's QPs shall prepare and endorse the necessary calculations, details and drawings and submit it to the S.O. and AC for checking. It is the responsibility of Contractor's QP to submit the drawings and calculations to BCA to obtain the approval and permit to commence work.

- (d) At the pipe rack interface listed in P3.5.6(a)(ii), C1A and C2A contractors will provide the H20 coupler rebars for the Contractor to connect. The Contractor's QP need to provide the underground reinforced concrete stump to support the pipe rack column including the associated excavation and backfilling works and need to ensure the pipe rack reaction forces is within the vertical and lateral capacity of the piles foundation as shown in the Drawings and provide the additional piles or other remedial measures if required. The Contractor's scope shall include all necessary coordination and liaison with C1A and C2A contractors for the connection works.
- (e) Pipe rack setting and section shown in BIM model for reference only, final setting out and structural steel member sizes shall refer to Contractor's QP design in accordance with the Specifications requirement.
- (f) Pipe rack coating and painting requirement shall be compatible with the C4A steelworks specifications stated in the General Notes in Volume 6. Minimum fire rating requirement as shown on the Drawings.

P3.5.7 Storm Water Drain Interface

- (a) The Contractor shall note that the storm water drains constructed under this Contract shall be connected to the major drains built by C1A contractor.
- (b) The connection at the interface shall follow the Rebated Joint detail shown in the Drawings.
- (c) The Contractor's scope shall include all necessary coordination and liaison with C1A contractor to make the drain connection and fill the rebated joint with cement mortar.

P3.5.8 Benching for Flow Channels

- (a) All flow channels shall be provided with mass concrete benching at the bottom of the channel.
- (b) The benching shall be provided at the bottom of the channel on both sides and the slope shall be at 45 degrees angle. The width of the benching at the base shall be 200 mm.
- (c) Benching size shall further be confirmed by the S.O. prior to the construction.

P3.5.9 Provision of Anchor Points for Maintenance Works

- (a) The Contractor shall propose location, design, supervision and installation of anchor points, fall arrest system and travel restraint system suited for future maintenance work to be carried out at height.
- (b) Such points shall be designed to act as anchoring points for Operations & Maintenance staff to hook their harness and to do maintenance in tanks.

- (c) The type and location shall be proposed by Contractor for various tanks and equipment for S.O.'s review and approval. Subsequently, these anchoring points shall be certified by the Contractor's PE and undergo mandatory testing as per MOM's requirements and guidelines.
- (d) The Contractor's rates and prices for supply of mechanical and electrical equipment, and construction of concrete tanks shall deem to include the provision of such anchors.

P3.5.10 Miscellaneous Work

The Contactor shall:

- (a) Propose and carry out trial mixes of concrete;
- (b) Provide PE endorsed design and obtain BCA approval as per authorities' requirements including but not limited to the following:
 - (i) Design and construct corbels and overhead runway beams for hoists and cranes;
 - (ii) Design and construct Mechanical, Electrical and ICA plinths, equipment support, equipment access, metal platforms/ access, metal staircase/ ladders, pipework support/ bridges, cable support/ bridges, covers and access panels etc.;
 - (iii) Pipe supports shall not restrict access to the pipes for maintenance. Details, calculations of proposed supports and checking of structural capacity of main structure shall be submitted to the S.O. for approval prior to installation of the pipework;
 - (iv) Design and construct various architectural elements and connection details to main structure, i.e. cladding, facade, roofing system, canopy, doors and windows frames, louvres, skylight, etc.;
 - (v) Design and construct handrail (barrier), bollards, vehicle impact guardrails, etc.;
 - (vi) Design and construct lintel beams, columns for door, windows, blockwall, walls to Contractor's PE details; and
 - (vii) Design and construct FRP covers and elements.

P3.6 WATER RETAINING STRUCTURES

P3.6.1 Control of Shrinkage

- (a) In water retaining structures and elsewhere as directed by the S.O., high workability, low water cement ratio mixes are required. To ensure suitable workability the Contractor may propose a mix design with an Approved-plasticising agent. The plasticiser shall be used in strict accordance with the manufacturer's instructions and General Specification G3 Civil and Building Works.
- (b) The Contractor has a responsibility to ensure that his sequence and method of construction do not compromise the integrity of the Permanent Works. To this end, the Contractor must ensure that his planned construction bay sizes do not exceed those specified in EC2-3 for the reinforcement detailed.
- (c) The Contractor shall submit a detailed method statement and a dated sequential programme showing his proposed layout and order of his concrete pours.

P3.6.2 Joints in Concrete

- (a) The Contractor shall submit details of all movement joints and including waterproofing details, as necessary, for the approval of the S.O.. No movement joints shall be used unless approved by the S.O.. Construction joints shall be used in locations as approved by the S.O. in accordance with the Contractor's approved method statement. All joints shall conform in every respect with the details shown in the Drawings. All construction joints in a water retaining section of the structure shall be made with a water stop within the concrete thickness. The Contractor shall submit a method statement for the approval of the S.O. of how high-quality compaction of concrete around the water bar will be achieved and how the water bar will be prevented from folding out of position during concrete pouring.

P3.6.3 Casting Walls

- (a) The Contractor shall adopt sequential casting of the walls for RC liquid retaining structures to avoid any end restraint condition.
- (b) The Contractor shall submit detailed shop drawings showing their intended construction sequence to S.O. for review and approval.
- (c) The closing wall panel subjected to end restraint may need additional reinforcement as required by S.O.. The Contractor shall provide and install the additional rebar as per S.O. instruction.
- (d) The walls shall be covered with a curing system as detailed in General Specification G3 Civil and Building Works and as approved by the S.O..

P3.6.4 Casting Sequence of Concrete Floors

- (a) The floor slabs shall be cast in a sequential way in which the sandwich casting of a new panel in between the panels which are already cast shall be avoided. The time, which shall elapse before casting any adjacent panel, shall be as approved by the S.O. in accordance with the Contractor's Approved method statement but shall not in any case be less than 3 days.

P3.6.5 Imperfections in Internal Concrete Surfaces

- (a) Imperfections in internal concrete surfaces that do not comply with the finish specified in General Specification G3 Civil and Building Works shall be rubbed down with carborundum stone, the resulting grit and dust being thoroughly washed off with clean water. After the rubbing down and as a separate operation a 1 to 1.5 cement and sand mixture shall be worked into the pores over the whole surface with a fine carborundum flint in such a manner that no more material is left on the concrete face than is necessary completely to fill the pores, so that an extremely smooth and dense surface shall be obtained, to the satisfaction of the S.O..

P3.6.6 Protection of Concrete Floors

- (a) No mechanical equipment/ machineries shall be permitted to run on finished floors and the floor area shall not be used without the consent of the S.O. for the storage or stacking of any materials. Steel scaffolding, if used for supporting formwork shall be erected on a suitable timber base, and the areas of floor immediately under any part of the structure, which is being concreted, shall be covered with sand, sacking, or sawdust, to prevent drips of grout adhering to the floor surface.

P3.6.7 Basement and Other Below Ground Structures

- (a) The internal surfaces of below ground structures shall be inspected for leakage after backfilling or during water testing of adjacent structures. Leakage shall be defined as any surface where water droplets, not caused by condensation, are visible, or any significant discoloration of the concrete caused by damp penetration that remains unimproved 7 days after the start of the test measurement period. Any areas of ingress of water shall be investigated and made good to prevent the inflow of water, whether from the ground or from the adjacent water retaining structure.

P3.7 BUILDING IN OF PIPEWORK AND OTHER ITEMS

P3.7.1 Reinforced Concrete

- (a) Pipework
 - (i) All pipes passing through water retaining structures, or pipes entering or emerging from structures below finished ground level shall have puddle flanges cast into the centre of the wall unless expressly instructed otherwise;

- (ii) Wherever possible, pipework to be provided under the Contract shall be built in as work proceeds in accordance with Contractor's PE details;
- (iii) Complete sections of pipework, including the items to be built into structures, shall be erected and checked by the Contractor for line and level before any items are built in. Casting in of individual pipes in isolation of the rest of the system will not normally be permitted, and any proposals in this respect shall be subject to the approval of the S.O.;
- (iv) The Contractor shall erect and fix securely in position all pipework, which is to be built in. Any supports shall be clear of the openings into which pipes are to be built, to allow access for installation of formwork and placing concrete;
- (v) The Contractor shall recheck the pipework alignment after erection of formwork and before placing concrete starts, to ensure that it has not moved;
- (vi) The Contractor shall be responsible for ensuring that the pipes are built in the correct position and true to line and level and special care shall be taken to ensure that concrete is adequately worked and placed around the pipe to secure a completely watertight joint;
- (vii) Particular care shall be taken to ensure that no pipe is displaced or disturbed in alignment whilst this operation is carried out. If pipes are found to have moved, the Contractor shall be required to cut out such pipes and re-fix them true to line and level at his own expense; and
- (viii) The Contractor may apply to the S.O. to install pipework or other cast in features using boxouts. The boxout edges shall be well scabbled and wedge shaped with the narrowest opening located on the water retaining side. The edges of the boxout shall be square to the face of the wall or slab for a minimum of 25 mm to avoid feather edges. The boxout shall have a 100 mm by 100 mm key groove to all sides at the centre of the wall or slab. Anti-crack steel shall be exposed at the corners of the boxout to improve the box out bond. The Contractor shall use a low water cement ratio ultra-high slump non-shrink concrete mix to cast in the boxout. The Contractor shall consider the use of 'Self-Compacting Concrete' to achieve the high level of compaction required. The Contractor shall submit for the approval of the S.O. the following details:
 - 1) Any proposed modifications to the steel reinforcement design including structural calculations, if required;
 - 2) The size and shape of the proposed box out;
 - 3) Methodology of ensuring good bond and key between new concrete used to fill boxout;
 - 4) Methodology for supporting pipe or fitting during casting in;

- 5) Mix design for concrete used to cast fitting in place. Mix shall ensure minimum shrinkage; and
 - 6) Methodology proposed to pour and compact concrete for casting fitting in place.
- (ix) The Contractor to take special care on grout loss on shutters around pipes and shall include in their rates for specialist sealing products to ensure grout loss in minimised. Contractor shall make special provisions to ensure that the concrete around the cast in pipe is adequately compacted. The Contractor shall make provision for specialist compaction equipment suitable for cast in pipes in their rates.
- (b) Masonry
- (i) Where pipework and other items are required to be built into masonry, wherever practical these should be built in as work proceeds. The S.O. may direct the Contractor to form box-outs during the masonry construction in accordance with details shown on the construction outline drawings issued to the Contractor by the S.O. or as Directed by the S.O. on Site for the subsequent building in of pipework or other items by the Contractor;
 - (ii) Precast concrete lintels shall be provided unless otherwise directed by the S.O.; and
 - (iii) Masonry used to build in around pipework and other items shall be of the same class and finish as the surrounding masonry, neatly cut to suit and with the finished masonry around the built-in item of the same class and finish as the surrounding masonry.

P3.8 OTHER STRUCTURES

- (a) General
- (i) All equipment and metalwork shall be raised a minimum of 30 mm above the finished floor level on a mortar, mass concrete or reinforced concrete plinth to avoid possible surface water ponding around equipment or metalwork base.
- (b) Window and Door Openings in Blockwall
- (i) The Contractor shall design, and provide reinforced concrete lintel beams for all door, window or other significant opening formed in brickwork walls as indicated in the Drawings or required for operation of the Works.

- (c) Rain Hoods to Doors or Openings
 - (i) The Contractor shall design and provide rain hoods to all external doorways and significant openings in walls unless expressly instructed otherwise. Rains hoods shall prevent most rain from hitting the doorway or opening and may be of concrete, FRP, aluminium or other Approved material. Details of proposed rain hoods shall be submitted to the S.O. for approval prior to placing order.
- (d) Drainage Sumps
 - (i) The Contractor shall provide a drainage sump in the base of all water retaining structures or structures that may become filled with water unless they have permanent drainage or have been expressly indicated not to be fitted with a drainage sump. The drainage sump shall be large enough to allow the temporary provision of a small drainage sump pump or drainage pump suction hose and shall be located below a suitable access point.
- (e) Pipework Supports
 - (i) All pipework fittings and valves laid above ground level and inside the building shall be adequately supported for their self-weight and the weight of the fluid they will contain. Pipe supports including plinths shall be designed by the Contractor and shall not restrict access to the pipes for maintenance. Details of proposed pipe supports shall be submitted to the S.O. for approval prior to installation of the pipework.
- (f) Road and Footpath Access
 - (i) Road access shall be provided for the following structures. The Contractor shall submit the road access plan to the S.O.'s for approval:
 - 1) Sidestream Nitrogen removal - WST02
 - (ii) Footpath access shall be provided to all structures, pipework, valve, flowmeter and washout chambers and to all doors and access points unless road access is already provided. Footpaths shall be normal duty precast concrete as indicated in the Drawings unless specified otherwise. Where slopes exceed 1 in 3 gradient the Contractor shall design and provide concrete steps to suit. Details of the location of steps and footpaths shall be submitted to the S.O. for approval.
- (g) Support for Provision of Mechanical, Electrical and ICA Equipment
 - (i) The Contractor shall ensure that civil and structural works shall provide support for all mechanical, electrical and ICA equipment provision as necessary. This may include but not be limited to:
 - 1) Coring for pipe, duct or cable passage through walls and slabs;

- 2) Plinths, upstands, pipe support, access platform, staircase and miscellaneous metalwork as required;
- 3) Chasing cable runs in concrete or brickwork and making good afterwards;
- 4) Demolition and disposal of minor structures obstructing equipment installation;
- 5) Cutting of roads and structures to allow installation of pipe or cable routes;
- 6) Provision of temporary or permanent water supply and drainage as required for Plant; and
- 7) This support work is deemed to be included in the Contract.

(h) Drainage

- (i) All areas of the structure shall wherever possible have gravity drainage. Where it is not possible to provide gravity drainage, a drainage sump shall be provided of suitable dimensions to contain a sump pump.
- (ii) The Contractor shall provide drop-inlet chambers and scuppers pipes for the roads under the Contract and the associated connections to the storm drains. Where road and drop-inlet chambers constructed by Other contractors and connected to C4B drains, the Contractor shall liaise with Other contractors to confirm the location of the scupper drain pipes. The Contractor shall provide necessary openings in the storm water drains for connection by Other contractors. The Contractor shall allow in his prices and rates for all above works.
- (iii) The Contractor shall provide storm drainage at the Sidestream Nitrogen Removal Facility as shown on the Drawings. The storm drainage layout provided in the Drawings is a notional plan. The Contractor shall coordinate with C1A for the connections to the sitewide drainage system.

(i) Vehicle Access

- (i) Openings for vehicle access shall be formed with reinforced concrete columns and lintel to support the surrounding brickwork. The columns either side of the access opening shall be protected with 6 mm thick galvanised mild steel panels fixed to the columns. The Contractor shall propose a suitable plate shape and fixing system for the approval of the S.O.. The protective plates shall be painted with high visibility paint.

- (ii) The equipment inside shall be protected from reversing vehicles by steel bollards bolted to the concrete floor and painted with high visibility paint. The bollards shall be minimum 230 mm dia. and of a sturdy construction and shall be able to be laid flat or removed simply to allow the removal of equipment as necessary. The Contractor shall propose a suitable bollard and anchoring system for the approval of the S.O..
- (iii) The Contractor shall fix a raised strip to the slab to act as a back wheel stop for reversing vehicles. The strip shall not be rounded, stand approximately 150 mm above the finished floor level and may be of concrete or alternative material proposed by the Contractor for the approval of the S.O.. The strip shall be robust enough to take repeated heavy vehicle impact without damage and shall be positioned on site to suit. The strip shall not be continuous across the width of the vehicle access bay but shall have a 1500 mm wide gap at the centre to allow access.
- (iv) The channel drain at the entrance shall be protected with a heavy-duty galvanised hot dipped steel grating. The grating shall be capable of taking heavy vehicle loads without damage and shall be painted black. The grating shall be flush with the finished road level, securely supported and bolted in place to prevent movement during vehicle passage. The securing bolts shall be removable to allow the grating to be lifted and shall be recessed below the top level of the grating. The entrance way shall be slightly raised to prevent the ingress of rainwater under the door.
- (j) Emergency Escape and Access Ladders
 - (i) The Contractor shall provide adequate access to fresh air and escape from the buildings and comply with the relevant codes. Ladders and platforms shall be installed where required for safe access and escape from plant and equipment. Locations shall comply with FSSD requirements.

P3.9 ROAD AND FOOTPATH WORKS

- (a) The Contractor shall provide the permanent roads, footpaths and any other road elements to LTA standards.
- (b) The permanent roads shall be installed after the completion of all yard piping and services under the roads. The permanent road shall be Type 1 flexible pavement roads in accordance with the LTA's prevailing Standard Details of Road Elements. The permanent roads shall be installed to a finished level of nominally 5 m SHD.
- (c) The Contractor shall note the top finished level of permanent storm water drains shall be 5.15 m SHD in general. Where drains cross the entrance to buildings, the road shall ramp down from the building floor level to the road edge and the slope of the ramp shall be suitably adjusted considering the top level of the drain to the approval of S.O.

- (d) The permanent roads shall be installed with road pavements, road kerbs, drop inlet chambers, scupper pipes, gratings, road markings as shown in the Drawings and in accordance with the Board and the LTA's prevailing Standard Details of Road Elements. The connection of scupper pipes to the permanent drainage system shall be carried out by the Contractor.
- (e) All permanent roads at the Sidestream Nitrogen Removal Facility shall be rigid pavement.
- (f) The Contractor shall provide cow grass turfing to all areas not covered by roads and footpaths. Turfed areas shall be sloped to ensure that these are free draining to an appropriate storm water drainage area and shall not have any localised depressions which may result in ponding of water.

P3.10 SUBMISSIONS AND COORDINATION

P3.10.1 Submittal Requirements

- (a) The S.O.'s approval shall be obtained prior to proceeding with any work at site. The following lists are the minimum that shall be submitted to the S.O. for approval:
 - (i) Programme for the Works, including any special requirements, to show how each affects the Contract;
 - (ii) Method statements as needed to demonstrate to other agencies or to the S.O. that the Contractor has taken into account the needs of any special requirements:
 - 1) ERSS installation works;
 - 2) Pile installation;
 - 3) Pile load tests and integrity test;
 - 4) Pipe installation work;
 - 5) Concreting works; and
 - 6) Structural steel works.
 - (iii) Precondition and postcondition survey of buildings, structures, utilities and services;
 - (iv) Drawings / shop drawings / As-built for the following:
 - 1) Setting out plans for piles;
 - 2) Setting out plans for buildings and underground structures;
 - 3) Precomputation plans for road and drains;

- 4) Setting out plans for the ERSS works;
 - 5) Plan and details for the ERSS work;
 - 6) Material proposal for ERSS;
 - 7) Shop drawings for RC work;
 - 8) Shop drawings for steelwork;
 - 9) Mill certificate for all steel elements required for the ERSS including local test reports for mechanical tensile test and material chemical analysis;
 - 10) Weld test and sample testing for weld test;
 - 11) Proposed materials and the relevant test reports for all permanent works;
 - 12) Shop drawings for pipework;
 - 13) Load Test Reports for the piling works;
 - 14) "As-built" plans for piling works;
 - 15) "As-built" plans for buildings and structures;
 - 16) "As-built" plans for external works;
 - 17) Site wide temporary drainage scheme; and
 - 18) Design and plans for ECM system.
- (v) Material proposal and catalogues:
- 1) Concrete trial mix proposal and mix design;
 - 2) Concrete trial mix results;
 - 3) RC piles catalogues;
 - 4) Premix for roads; and
 - 5) Mill certificate for all steel elements, temporary or permanent, including local test reports for mechanical tensile test and material chemical analysis.
- (b) Contractor shall carry out QC testing and submit QC testing records including but not limited to the following:
- (i) Concrete cubes;
 - (ii) Concrete trial mix proposal and mix design;

- (iii) Concrete trial mix results;
 - (iv) Reinforcing bars, tensile and bend test;
 - (v) Structural steel tensile test, chemical test, charpy impact test and other relevant material testing;
 - (vi) Pipe material and mechanical testing;
 - (vii) Earth compaction testing and load test reports for the piling works;
 - (viii) Weld test and sample testing for weld test;
 - (ix) Tests for premix; and
 - (x) Other materials testing and relevant QC test reports for all permanent works.
- (c) Copies of meeting minutes for coordination meetings.
- (d) All submissions shall be made in soft and hard copy format. The Contractor shall submit a minimum of 2 hard copies, and one soft copy. Where submission involves review by other parties, such as contractual matters by QS, design matters by C&S designer, the Contractor shall make additional copies as required. The S.O. reserves the right to request additional copies of submission from time to time and the Contractor shall make allowance in his pricing for the provision of additional copies.
- (e) The Contractor shall make timely submission for the above listed items or any other items requiring S.O.'s approval. Contractor shall allow minimum 2 weeks review period for the S.O. to respond to the submission. The Contractor shall be required to make the submission 4 weeks in advance if it is critical to the work progress or if it involves multiple reviews stages to secure the required approvals.
- (f) The Contractor shall promptly reply with resubmission in response to S.O.'s comments. The maximum response period shall be 2 weeks.
- (g) The Contractor shall track all submission and be responsible for managing the submission and obtaining the necessary approvals in time.
- (h) The Contractor shall submit their Quality Assurance & Quality Control Plan for all the above works to S.O for approval upon award of the Contract.

P3.10.2 Works Affecting Existing Building and Structures

- (a) The Contractor shall note that the Works will be carried out in close proximity to the works by Other contractors. The Sidestream Nitrogen Removal Facility (WST02) construction works is adjacent to the C1A and C3A existing piles which are supporting the C3B1 and C4B pipe rack. The Contractor's QP needs to study the structural scheme to accommodate current site constraint and carry out the impact assessment and are deemed included in the Contract Sum.

- (b) The Contractor shall submit their proposed instrumentation for monitoring plan to monitor vibration, soil settlement and movement etc. due to his works. If the readings exceed the permitted levels, the Contractor shall inform the S.O. on the remedial measures that will need to be taken.
- (c) The Contractor shall work with the S.O. to compile an emergency plan for any unforeseen situation that concerns the existing services and the safety of users e.g. settlement of ground, cracking of structures, unstable supports, etc.
- (d) The Contractor shall take all safety precautions while work is in progress. The Contractor shall ensure at all times that its construction activities do not pose a safety hazard to their staff or Other contractors. All safety precautions must be taken while work is in progress and the S.O. shall be consulted if any work should have an impact on the safety of Other contractors.
- (e) The Contractor shall provide the schedule of works to the Other contractors at least two weeks before commencement of construction works within their compound. Thereafter, updates on progress of works and regular checks to structures for e.g. cracks, soil settlement, ground upheaval, depression or cavitation, shall be carried out. The Contractor shall seek regular feedback from the Other contractors pertaining to his construction works.
- (f) Noise disturbance shall be kept within permitted limits specified by National Environment Agency. The Contractor shall regularly monitor the noise level at the site. The Contractor shall comply with the requirements under the Workplace Safety and Health (Noise) Regulation 2011.
- (g) The Contractor shall monitor and ensure that the dust generated from the Works do not affect the air quality within the site and surrounding compound and are kept within approved levels set by the National Environment Agency (NEA).
- (h) The Contractor shall put up adequate notices to inform personnel for “out-of-bound” areas. Where necessary, physical barriers shall be provided.
- (i) The Contractor shall ensure that there is no disruption to underground services and if any, the services must be restored immediately. The Contractor shall provide temporary services if disruption cannot be fully restored.
- (j) The Contractor shall monitor and ensure that structural integrity of the existing structure is maintained at all times. In addition, the Contractor shall ascertain and ensure there is no ground movement or building movement in the site premises.
- (k) In the event that any urgent matters arise, the Contractor and his subcontractors must contact the S.O. immediately.
- (l) All costs pertaining to repair works are to be borne by the Contractor and/or its relevant subcontractors.
- (m) The Contractor shall include such coordination and necessary protection works for the entire completion of proposed works in his Contract Sum.

P3.10.3 Requirements for Coordination and Liaison with Other contractors, Government Agencies, Statutory Boards, Utilities Companies, etc.

- (a) The Contractor shall note that a high degree of cooperation will be required in programming elements of the Works, which interface with other contracts, particularly C1A, C2A, C3A, C3B1, C4A and C6B. The Contractor shall allow for minor variations in scheduling work activities and shall allow for working closely with the S.O. and Other contractors in this regard.
- (b) The Contractor shall be responsible for the provision and maintenance of overall site hoarding for the work site for the entire Contract duration including manning the entrance gate with security guard at all times.
- (c) The Contractor shall allow in his prices and rates for all co-ordination works including monitoring and liaison with Other contractors, Government Agencies, Statutory Boards, Utilities Companies, etc., on interfacing and/or working along the water pipelines, and at temporary shafts and chamber locations.

P3.10.4 Survey Coordination

- (a) The Contractor shall coordinate with Other contractors to mutually agree on survey control for the interfaces between his Works and the adjoining contracts.

P3.10.5 Sharing of Work Access

- (a) The Contractor shall allow the sharing of site access with any Other contractors appointed by the Board.
- (b) There will be times when Other contractors may require the use of common access, site entrance, internal temporary work access, etc. The Contractor shall grant access to Other contractors employed by the Board that require the passing through of his site to perform their required work.

P3.10.6 Existing Utilities

- (a) The existing utilities drawings are given to the tenderers, but it is not exhaustive. Contractor is required to purchase the existing utilities drawings from the Utility Agencies and carry out the manual detection at site by trial trenches to verify the location of utilities. The cable detection shall also be carried out by the licensed cable detection worker at site prior to any construction works. There may be utilities which are not shown in the purchased drawings but may be present at site which needed to be supported, protected or diverted by the Contractor to facilitate the construction works. All the associated costs for the works are deemed to have been included in Contract Sum.

P3.11 EARTHWORKS

P3.11.1 General Requirements for Earthworks

- (a) An ordnance survey of the Site shall be carried out prior to the commencement of the earthworks. As the earthworks progress, the Contractor shall perform additional ordnance surveys at various depths (and ordnance removal if necessary) to ensure the safety of the earthworks operation.
- (b) The Contractor shall take precautions to minimise dust pollution associated with earth movement, fill and compaction.
- (c) All bare surfaces including the fill areas shall be protected and covered with erosion control blanket. All elements of the erosion control systems shall be installed in conjunction with the earthworks and shall be maintained continuously as the earthworks progress.
- (d) During the earthworks operation, the Contractor shall monitor the conditions of the roads closely to ensure that the roads and utilities are not damaged. Dust control and wheel wash measures shall be maintained consistently.
- (e) Topographical survey plans are included in the tender documents. The Contractor may perform his own additional survey as he deems necessary. Unless otherwise informed in writing by the Contractor prior to the start of earthwork activities, the topographical survey plan shall be the record survey for assessment of earthwork quantities.
- (f) The Contractor shall coordinate with the Other contractors to ensure that his works does not negatively impact the ERSS and works installed by Other contractors.

P3.11.2 Earthworks to support the Permanent Works

- (a) The Contractor shall excavate from the existing ground levels to the required formation levels of the work areas to allow construction of all proposed facilities under this Contract and shall fill the areas to the final formation level of 5 m SHD. The Contractor shall refer to Volume 6 – Anticipated Final Levels for the existing ground levels.
- (b) The Contractor shall be responsible for the earthworks in his working areas. The fill shall be carried out with either material from the surcharge, excavated material from the Site or importing materials from the Contractor's own source.
- (c) The Contractor shall note that the fill to the final formation level of 5 m SHD will be permitted only when structures adjacent to the fill area have been completed up to 5 m SHD. The Contractor shall allow for this in the Programme for the Works, which will need to be updated to reflect changes to other TWRP contractors' programmes from time to time.

- (d) The Contractor shall submit the earthworks plan and sequence for the proposed filling of the areas under his charge to 5 m SHD to the S.O. for approval prior to the commencement of earthworks.
- (e) The Contractor shall carry out the field and laboratory tests in accordance with the Specifications to ensure that material that is used as fill material complies with the requirement for fill as specified. The material from the excavation works for the proposed facilities that is found to be unsuitable as fill material shall be disposed of at an approved offsite dumping ground.

P3.11.3 Fill and Backfill

- (a) The Contractor shall be responsible for ensuring that adequate materials are available on Site for the fill and backfilling of his works to meet the schedule. The Contractor shall import approved material from outside the Site for the fill and backfilling of his works as necessary.
- (b) All backfilling shall be graded to ensure drainage provisions on the Site are maintained at all times.
- (c) The Contractor shall coordinate backfill activities with the Other contractors. The Contractor shall develop the backfilling plans in conjunction with the S.O. and the Other contractors.
- (d) The Contractor shall exercise caution to prevent damage to utilities and other site facilities during the backfill operations. The Contractor shall be responsible in making good any such damages caused by his works and shall bear all costs incurred.
- (e) During the backfill operation, the Contractor shall be responsible for exercising all precautionary measures to protect waterproofing material applied on the structures from any damages. The Contractor shall be responsible for liaising with the same specialist waterproofing contractor in making good any such damages caused by his works and shall bear all costs incurred.

P3.11.4 Stockpiling

- (a) The Contractor may stock pile materials temporarily on Site within his working area and for the purposes of backfilling. The stock pile proposal including the areas and levels of the temporary stockpile shall be submitted to the S.O. for approval prior to stock piling.
- (b) The Contractor shall note that there are limited staging areas on Site and all excavated material shall be disposed offsite without stock piling if the no stockpiling area is identified within the Contractor's working or staging area.

- (c) The Contractor shall maintain the stockpile to ensure it is not contaminated or mixed with unsuitable materials. Where stockpiles become contaminated or mixed, and the material is intended for backfill operations, the Contractor shall be responsible either for separating the different types of soils to the complete satisfaction of the S.O. or, where separation is not possible, removing the contaminated materials from the Site and replacing them with suitable material at the Contractor's cost.
- (d) Stock piles must be maintained with appropriate slopes and earth control measures. All stockpiles shall be protected and covered with erosion control blanket.

P3.11.5 Disposal of Unwanted Materials

- (a) All surplus spoil not required for backfilling or filling purposes shall be carried away immediately after excavation / treatment to the approved dumping ground. The Contractor is not allowed to deposit surplus material on any State or private land.
- (b) The Contractor shall be responsible for finding on his own and at his own cost, suitable and approved dumping grounds for the disposal of all surplus and unwanted materials excavated or dredged, debris, spoils and all other construction waste materials, resulting from the site clearance, dredging, excavation and all works under this Contract.
- (c) The Contractor shall note that illegal dumping of unwanted building debris, excavated materials, dredged material, etc. on any vacant plot of land, roadside or drains is an offence.
- (d) No unwanted excavated materials shall be deposited in any area other than the approved dumping grounds provided by the Contractor. The Contractor shall indemnify the Board against any claim resulting from the Contractor's default in this respect. In addition, the Contractor shall, at his own cost, carry out remedial works to the satisfaction of the S.O. or the relevant authorities having jurisdiction over the area. If the Contractor fails to carry out the remedial works, the S.O. may employ workers, machinery and all else necessary for the removal and proper disposal of such dumping and all costs incurred thereof shall be deducted from any monies due or to be due to the Contractor.
- (e) Excavated materials and waste removed from the Site shall be controlled and disposed off-site in a manner that meets all environmental, erosion and siltation regulations. All such materials shall first be segregated prior to removal from Site in bins for disposal at the Contractor's own dumping grounds.
- (f) The Contractor shall comply with all aspects of safety, environmental protection and security requirements in regard to spoil disposal as prescribed in the Contract and as may be required by the authorities or under any enactment, order, ruling or regulation. The Contractor shall always keep all access required for work by others free of all debris, excavated materials and any other obstructions.

- (g) The Contractor shall note the difficulties of access to and obtaining approvals for dumping of earth or other spoils at established government owned dumping grounds. Due to the heavy traffic and congestion, there is a limitation in the quantity of soil that can be transported daily.
- (h) The Contractor shall take note the strict criteria and standards set on the type of soil that can be disposed at the established dumping grounds. Soil of poor quality, soft soil, soil with heavy metals is not permitted to be disposed at these established dumping grounds. The Contractor shall refer to the latest requirements from the approved dumping grounds, for pricing of his Tender. Under such circumstances, the Contractor shall seek alternative arrangements for disposal of surplus earth from the Works, which could include, but not limited to, earth filling at other suitable on-going projects, treating the soil to a quality suitable for disposal, etc.
- (i) Where the Contractor proposes to treat the earth prior to disposal, the Contractor shall first engage experienced Environmental Specialists or Experts to identify suitable treatment and disposal options. The Contractor shall manage the whole process from treatment, waste disposal and sourcing for any suitable dumping ground that may accept the soil to be disposed of.
- (j) The Contractor shall include in his Contract Sum the hauling and disposal of the debris and excavated material at the Contractor's own dumping grounds, including spreading the spoil at the point of disposal.
- (k) The Contract Sum shall also include the setting out of launching ramp at the foreshore of the Site where this is necessary and the provision of barges, if the Contractor's proposed dumping ground is located offshore.
- (l) At the start of the Contract:
 - (i) The Contractor shall declare the location of the disposal site for surplus earth or excavated materials. The Contractor may choose to dispose of them via the Changi Staging Ground Facility off Tanah Merah Coast Road. If the Contractor intends to dispose of excavated soil/materials to his own dumping grounds, he shall submit to the S.O. letters of consent from the management/owners of the disposal sites and, if the ground level of the disposal site would be raised, also approvals from the URA, BCA and relevant authorities.
 - (ii) Where the Contractor proposes to dispose of excavated materials at a private site, he shall produce photos showing the site before and after the disposal operation.
 - (iii) The Contractor shall clearly and prominently demarcate on site the boundary of the disposal site such that drivers transporting materials to the disposal site will not unload materials outside the disposal site by mistake.
 - (iv) The Contractor shall carry out all necessary soil sampling and testing of excavated material to the satisfaction of the operators of the approved disposal sites. All costs associated with this are deemed to be included in the Contractor's Contract Sum.

- (v) The Contractor is to note that it is entirely his responsibility to ensure that the classification of the material disposed at the approved disposal sites meet the operator's classification and guidelines. Any non-compliance on the part of the Contractor with the operator's requirements which may result in the Contractor being barred from using the disposal facility shall not absolve the Contractor of any responsibility towards carting away all excess excavated material off the project Site and the Contractor shall not be entitled to any additional cost or extension of time as a result.
- (m) During the progress of the Works:
 - (i) The Contractor shall seek the S.O.'s prior approval for any change of disposal site during the Contract.
 - (ii) Disposal of earth shall be carried out only during daylight hours. If the Contractor wants to dispose of excavated earth outside the daylight hours he is to seek prior written approval from the disposal facility operator and the S.O.. If disposal works are carried out outside daylight hours, the Contractor shall submit disposal chits / tickets duly endorsed by authorised persons at the disposal facility to the S.O. by 8:30 a.m. the following morning.
 - (iii) If the volume of earth to be disposed of from any site exceeds 1000 m³, and if the earth is to be disposed of at a private site, the Contractor shall employ a person, whose personal particulars and specimen signature shall be lodged with the S.O., to be stationed at the disposal site to acknowledge receipt of excavated earth. The Contractor shall brief that person thoroughly and obtain a written statement from him to the effect that he shall fully understand that he will be liable for aiding and abetting illegal dumping if he makes false acknowledgements.
 - (iv) The Contractor shall submit the name of the haulage contractor and a list of registration numbers of the trucks that will be used in the transportation of excavated materials. All trucks shall have an up-to-date log book in which the driver shall enter all relevant details of trips made. The Contractor shall inform the S.O. immediately of any change in the haulage contractor and registration numbers of trucks that will be used for disposal of excavated material, during the Contract.
 - (v) The Contractor shall put in place a ticketing system to monitor the removal of excavated materials from the Site and the receipt of the materials at the disposal site. The tickets shall be serially numbered and contain at least the following information:
 - 1) Date and time of removal from site;
 - 2) Name and ID of driver;
 - 3) Truck registration number;
 - 4) Location of disposal site;

- 5) Date and time of receipt of earth;
 - 6) Contractor's stamp and signature at the beginning of the journey; and
 - 7) If the disposal site is a private site, the Contractor's stamp and signature of the Contractor's authorised person at the disposal site.
- (vi) If the disposal site is one that is managed by the BCA, MSE or HDB the original copy of the weighbridge docket shall be submitted to the S.O..
 - (vii) A copy of the ticket shall be submitted to the S.O. immediately after each trip is made for disposal of excavated materials.
 - (viii) If the disposal site is a private site, the Contractor shall submit photographs monthly showing activities connected with the disposal of excavated materials to the S.O..
 - (ix) The Contractor shall provide transport for the S.O. and his representatives to the disposal sites as and when required to conduct audit check on the disposal operation.
 - (x) The Contractor is deemed to have allowed for all contingencies, risks and necessary costs and fees for the disposal of excavated materials and construction waste and for complying with all requirements stated in this clause in the pricing of this tender. No claim by the Contractor relating to the disposal of unwanted materials will be entertained by the Board.
 - (xi) The Contractor shall inspect, test and classify all spoil for proper disposal.
 - (xii) Marine clay (if any) shall be disposed directly to the approved disposal site. For materials that are hauled to the approved disposal site, the Contractor shall be responsible for coordinating procedural matters with the authorities for the disposal of excavated materials.
 - (xiii) The Contractor shall note that ground improvement work using vibro-flotation and perforated vertical drains with surcharge had been carried out on the TVB. The Contractor shall be responsible for ensuring that the soil disposed to the approved disposal site is free of debris.
 - (xiv) The Contractor shall dispose of all approved material (which includes good earth, soft clay and unsuitable earth as defined in the General Specifications) excavated and treated (as necessary) to the approved dumping ground.
 - (xv) The Contractor shall prepare all necessary information as required by the Board and authorities/ agencies managing the dumping ground to facilitate the Contractor's application for the disposal of the material at the approved dumping ground. Prior to obtaining necessary approvals, the Contractor is prohibited from disposing all material. Application requirements are specified below.

- (xvi) The Contractor shall also note that the S.O.'s approval to the Contractor's proposed haulage contractor to dispose soil at the approval dumping ground is provisional and subject to the authorities' / agencies' approval of the Contractor's application.
- (xvii) The Contractor shall not assume that the rate of progress of disposal at the approved disposal site will match his excavation rate. The Contractor shall make allowance for delays for off-loading, facility closure and other restrictions imposed by the authorities / agencies. The Board shall not be responsible for any delays or shortfall of the loading points or dumping facilities or its operation by the authorities / agencies managing the approved dumping ground.
- (xviii) The Contractor shall ensure that all the earth truck operators observe all traffic arrangements at the approval dumping ground. Upon the S.O.'s direction, the Contractor shall remove any earth truck operator who refuses to observe the required traffic arrangements, operational rules and requirements or causes mischief.
- (xix) The Contractor shall take all reasonable measures to avoid mixing different types of spoil, unless they are to be disposed of at the same location within the same facility. If the S.O.'s staff and the Contractor cannot agree on the classification of material for disposal, the matter shall be referred to the S.O. for his decision, which shall be final and binding.
- (n) For all materials leaving the Site by truck, the Contractor shall comply with all regulations regarding excessive loading, covering of loads, cleaning of wheels and truck bodies, cleaning roads and speed limits.
- (o) The Contractor shall maintain complete records of all disposal materials regarding the destination and delivery records.

P3.12 MEASUREMENT OF ROCK EXCAVATION

- (a) Rock excavation for the purposes of measurement shall be defined as any excavation in Jurong Formation of classification S2 or stronger, which cannot be excavated without special excavation methods such as drilling (by hand or with heavy machinery) or blasting with explosives. The Contractor shall provide evidence to support the need for special excavation methods to the S.O. for approval.

P3.13 APPENDIX

- (a) APPENDIX P3A – Exposure for Concrete Structures

END OF SECTION

P3A EXPOSURE FOR CONCRETE STRUCTURES

P3A.1 Exposure for Concrete Structures

Process Fluid Segments	Coverage Type*	Corrosion Protection System
Plant Drain manholes	1	PVC Lining
Sidestream Nitrogen Removal – Liquid Retaining Structures	2	PVC Lining
Chemical Bunds	3	Epoxy System (Novolac, Bisphenol F or Equivalent)

Notes:

1. Table shall be read in conjunction with Drawing sheet no. 131101-WSW41-DR-S-5001.
* Proposed options during construction are subjected to S.O. approval.